Week 3

Introduction

In the rapidly evolving landscape of modern power systems, efficient and reliable power conversion is not just an advantage—it's a necessity. From the burgeoning electric vehicle charging infrastructure to sophisticated industrial power supplies, the ability to convert AC power to DC with high efficiency and minimal grid disturbance is paramount. This week, we delve into the intricate world of the single-phase boost Power Factor Correction (PFC) converter, a fundamental building block in achieving these goals. Building upon foundational knowledge, we transition from understanding basic operation to the critical practicalities of component sizing and, crucially, the sophisticated control strategies required to make these converters perform optimally.

This journey is essential for any engineer aiming to design robust and compliant power electronics. We will explore how to meticulously select passive components like inductors and capacitors, and define the voltage and current ratings for active switching devices and diodes, ensuring the hardware can withstand operational stresses. Subsequently, we will pivot to the intelligence behind the hardware: closed-loop control. This involves understanding how to dynamically manage the converter's behavior to maintain a stable output voltage and draw a unity power factor current from the AC source, thereby maximizing efficiency and adhering to stringent power quality standards. By the end of this week, you will have a comprehensive understanding of both the physical design and the intelligent control mechanisms that underpin high-performance PFC converters.

Core Concepts

Term	Definition	Significance
`Unity Power Factor (UPF)`	A condition where the current drawn from the AC source is perfectly in phase with the voltage, minimizing reactive power and maximizing real power transfer.	Essential for efficient power utilization, compliance with grid regulations, reducing losses, and improving system performance.
`Double Line Frequency Ripple`	A significant ripple component in the output voltage or current that occurs at twice	This harmonic significantly impacts the sizing of the output capacitor, often leading to

Term	Definition	Significance
	the input line frequency (e.g., 100 Hz for a 50 Hz input).	large component dimensions in single-phase converters.
`RMS Current`	The root mean square value of a time-varying current, representing its heating effect equivalent to a DC current.	Fundamental for selecting appropriate power switches, diodes, and inductor wire, as it dictates their thermal stress and power dissipation capabilities.
`Duty Ratio (D)`	The fraction of a switching period during which a power switch is in its ON state. In PFC, it varies to maintain a constant output voltage and shape the input current.	Crucial for controlling the converter's output voltage and shaping the input current to achieve power factor correction.
`Average Current Control`	A control strategy that regulates the average inductor current by adjusting the duty ratio of the switches.	A common and effective method for PFC converters, ensuring the input current follows the input voltage waveform for UPF.
`Small-Signal Modeling`	A technique used to derive transfer functions by linearizing the converter's non-linear equations around an operating point.	Crucial for designing stable and responsive controllers by providing a mathematical representation of the plant's dynamic behavior.
`State Variables`	A minimal set of variables that completely describe the dynamic state of a system at any given time, typically inductor current and capacitor voltage.	Form the basis for state-space modeling, allowing for a comprehensive mathematical description of the converter's behavior.
`Gi(s)`	A transfer function representing how the inductor current in the boost PFC converter responds to changes	Crucial for designing the inner current control loop, ensuring

Term	Definition	Significance
	in the duty ratio of its switches.	the actual current tracks the reference.

Detailed Analysis

1. Component Sizing for Single-Phase Boost PFC Converters

The design of a single-phase boost PFC converter begins with the meticulous sizing of its passive and active components, as discussed in **Single-phase Boost PFC Converter - III** and **Single-phase Boost PFC Converter - IV**. The converter typically consists of a full-bridge diode rectifier followed by a boost converter stage. The rectifier converts the AC input voltage ($V_s = V_{s,pk} \sin(\omega t)$) into a rectified DC voltage (V_d), which then feeds the boost stage. A critical operational constraint for the boost converter is that its output voltage (V_o) must always be greater than the peak input AC voltage ($V_{s,pk}$). For example, for a 230V RMS input, $V_{s,pk}$ is approximately 325V, necessitating an output V_o of 400V or higher.

- **Inductor Sizing (L)**: The inductor value is determined by the desired maximum current ripple (typically 10% of $I_{s,pk}$), the output voltage (V_o), and the switching frequency (f_s). A higher switching frequency generally allows for a smaller inductance value. The RMS current through the inductor ($I_{L,rms}$) is approximately $I_{s,pk} / \sqrt{2}$, which is crucial for selecting the appropriate wire gauge to prevent excessive heating.
- **Capacitor Sizing (C)**: Sizing the output capacitor is critical for maintaining the output voltage within an allowable ripple (ΔV_o). A significant challenge in single-phase converters is the presence of a "double line frequency" ripple on the output capacitor voltage. This ripple occurs at twice the AC line frequency (e.g., 100 Hz for a 50 Hz line). The formula for capacitance is derived from power balance and the capacitor current ripple:
  $C = P_L / (2\pi f_s * \Delta V_o * V_o)$
  This low ripple frequency necessitates a significantly larger capacitance value compared to converters operating at higher switching frequencies, often making the output capacitor a bottleneck in terms of size and cost. The capacitor's RMS current ($I_{C,rms}$) is also vital for selection, as it dictates the capacitor's ripple current rating and directly impacts its lifespan.
- **Switch Ratings (S1 and S2)**: Both main switches (S1 and S2) must be rated to block the full output voltage (V_o) when they are in the OFF state. A safety margin of 40% to 60% is typically applied (e.g., $1.4 * V_o$). The RMS current for S1 ($I_{S1,rms}$) and S2 ($I_{S2,rms}$) are calculated by integrating the

square of the current over a half-line cycle, considering the time-varying duty ratio and input current. These RMS values are crucial for selecting appropriate devices (MOSFETs or IGBTs) based on their thermal capabilities. For IGBTs, the average current rating for S2 (I_S2_avg) is also a key selection criterion.

- **Rectifier Diode Ratings (D1-D4)**: The diodes in the input rectifier stage must withstand a peak reverse voltage (VRRM) at least equal to the peak input AC voltage (V_s,pk). Each diode conducts for half of the line cycle, so their forward RMS current (I_D_rms = I_s,pk / 2) and average current (I_D_avg = I_s,pk / π) must be calculated for proper selection.

Introduction

Once the physical components are sized, the next step is to implement a control strategy to ensure the converter operates as intended. As introduced in **Closed Loop Control of Single-phase Boost PFC Converter**, the primary objectives of closed-loop control are:
* **Output Voltage Regulation**: Maintaining a constant DC output voltage (V_o) despite input voltage fluctuations or load changes.
* **Unity Power Factor Current Draw**: Ensuring the converter draws a current from the AC source that is in phase with the input voltage.

The **duty ratio (D)** of the active switches (S1 and S2) is the primary control variable. A common strategy is **Average Current Control**, which employs a hierarchical structure:
* **Outer Voltage Control Loop**: This loop compares the desired output voltage (V_o reference) with the sensed actual output voltage (V_o feedback). A voltage controller (e.g., PI controller) processes the error and generates a signal representing the required magnitude of the inductor current (I_L,pk) to correct the voltage. A **saturation block** limits this current demand to protect components. This I_L,pk is then multiplied by the modulus of the unit sine function (modulus of sin(ωt), derived from the input AC voltage) to create the **I_L reference**, ensuring the input current tracks the input voltage waveform for UPF.
* **Inner Current Control Loop**: This loop compares the I_L reference with the sensed actual inductor current (I_L feedback). A current controller processes this error and adjusts the duty ratio (D) to ensure the actual inductor current closely follows its reference. This duty ratio is then fed to a modulator block to generate PWM signals for the switches.

To design these controllers effectively, understanding the converter's dynamic behavior through **small-signal modeling** and **transfer functions** (Gi(s) and Gv(s)) is crucial.

3. Deriving the Average Large Signal Model using State-Space

To analyze the converter's dynamics, we first develop an **average large signal model** using state-space representation, as detailed in **Closed Loop Control of Single-phase Boost PFC Converter - II**. This model averages the converter's behavior over a switching period, simplifying analysis by removing high-frequency ripple. The system's **state variables** are the inductor current (`iL`) and the capacitor voltage (`vc`). The inputs are the duty ratio (`D`) and the rectified input voltage (`|vs|`).

The converter operates in two distinct modes within a switching period (T_s):
* **DTs Period (S1 ON, S2 OFF)**:
* `diL/dt = |vs| / L`
* `dVc/dt = -vc / (RL * C)`
In state-space form, `x_dot = A1 * x + B1 * |vs|` , where `x = [iL; vc]` .
* **(1-D)Ts Period (S1 OFF, S2 ON)**:
* `diL/dt = (|vs| - vc) / L`
* `dVc/dt = (iL - vc / RL) / C`
In state-space form, `x_dot = A2 * x + B2 * |vs|` .

The **average large signal model** is obtained by weighting these two modes by their durations:
`x_dot_avg = D * (A1 * x + B1 * |vs|) + (1 - D) * (A2 * x + B2 * |vs|)`
This simplifies to:
`x_dot_avg = A * x + B * |vs|`
Where `A = D * A1 + (1 - D) * A2` and `B = D * B1 + (1 - D) * B2` .
The resulting average A matrix is:
`A = [[0, -(1-D)/L], [(1-D)/C, -1/(RL*C)]]`
And the B matrix is:
`B = [[1/L], [0]]`
This average state-space model forms the foundation for deriving small-signal transfer functions.

4. Small-Signal Modeling and Derivation of Gi(s)

Building on the average large signal model, **Closed Loop Control of Single-phase Boost PFC Converter - III** focuses on deriving the small-signal model and, specifically, the **Gi(s)** transfer function. Since power converters are non-linear, **linearization** is performed by introducing small perturbations around a steady-state operating point. Each variable is expressed as its steady-state value plus a small, time-varying perturbation (e.g., `D = D + d_tilde`). Products of small perturbation terms are considered negligible.

After substituting these perturbed variables into the average state-space equations and separating steady-state from small-signal terms, the linearized small-signal state equations are obtained. These equations describe how small perturbations in duty ratio (`d_tilde`) and input voltage (`Vs_tilde`) affect small perturbations in inductor current (`iL_tilde`) and capacitor voltage (`vc_tilde`).

To obtain the transfer functions, the linearized time-domain equations are transformed into the s-domain using the Laplace transform. For the current loop transfer function, **Gi(s) = iL(s) / d(s)**, we assume no perturbation in the input voltage (`Vs_tilde(s) = 0`). The detailed derivation involves inverting the `(sI - A)` matrix and performing matrix multiplications. The final derived form of `Gi(s)` is:
`Gi(s) = iL(s) / d(s) = [Vs / (RL * (1 - D)^3)] * [(RLC * S + 2) / (S^2 * LC / (1 - D)^2 + S * L / (RL * (1 - D)^2) + 1)]`
This second-order transfer function, with a single zero, provides critical insights into the dynamic response of the inductor current to changes in the duty ratio, which is essential for designing a stable and responsive inner current controller. The next step in control design would be to derive `Gv(s)` , relating output voltage to inductor current, to complete the picture for the outer voltage loop.

Key Takeaways

- [ ] The output voltage of a single-phase boost PFC converter must always be greater than the peak input AC voltage.
- [ ] The double line frequency ripple on the output capacitor necessitates a large capacitance value, posing a significant design challenge for single-phase converters.
- [ ] Component sizing for switches and diodes requires careful calculation of both voltage ratings (with safety margins) and RMS/average current ratings for thermal management.
- [ ] Closed-loop control is essential for achieving both regulated output voltage and unity power factor in PFC converters.
- [ ] Average current control uses an outer voltage loop to generate a current reference and an inner current loop to track it, shaping the input current.
- [ ] Small-signal modeling, involving linearization around an operating point, is crucial for deriving transfer functions of the inherently non-linear converter.
- [ ] State-space representation provides a structured mathematical framework for modeling the converter's dynamics during its different switching states.

- [ ] The `Gi(s)` transfer function, relating inductor current to duty ratio, is fundamental for designing the inner current control loop.

Conclusion

This week's exploration has taken us through the critical stages of designing a single-phase boost PFC converter, from the foundational task of sizing its components to the sophisticated realm of closed-loop control. We've seen how meticulous calculations for inductors, capacitors, switches, and diodes ensure the physical integrity and thermal performance of the converter, addressing challenges like the pervasive double line frequency ripple. More importantly, we've delved into the intelligence that transforms these components into a high-performance system: closed-loop control.

Understanding the average current control scheme, with its nested voltage and current loops, is paramount for achieving the dual objectives of stable output voltage and unity power factor. The rigorous process of small-signal modeling and the derivation of transfer functions like `Gi(s)` provide the mathematical blueprint for designing robust and responsive controllers. These principles are not merely theoretical exercises; they are the bedrock for developing advanced charging infrastructure, efficient power supplies, and reliable energy conversion systems that meet the demanding requirements of modern applications. As we move forward, the ability to translate these analytical insights into practical controller designs will be key to engineering truly intelligent and efficient power electronics solutions.